

Getting to School: Survey Data

AP Statistics · Unit 1 · Topic 1.2 · TEACHER KEY

CED TETHER

- **SKILL: 2.A** — Describe data presented numerically or graphically.
- **ENDURING UNDERSTANDING: VAR-1** Given that variation may be random or not, conclusions are uncertain.
- **VAR-1.B** Identify variables in a set of data. [Skill 2.A]
- **VAR-1.B.1** A variable is a characteristic that changes from one individual to another.
- **VAR-1.C** Classify types of variables. [Skill 2.A]
- **VAR-1.C.1** A categorical variable takes on values that are category names or group labels.
- **VAR-1.C.2** A quantitative variable is one that takes on numerical values for a measured or counted quantity.

Essential question: When is a number a quantity, and when is it just a label?

DOK-3 skill: 4.B · **FRQ pattern:** number-coded-variable-and-meaningless-summary

DOK rationale: Part (c) asks the student to adjudicate two students' classifications of number-coded variables, justify each with the definition, and predict what specific calculation becomes meaningless under the wrong choice; both classifications of the rating can be defended, so the reasoning, not the label, is what is scored.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- What does a zip code measure? What does it count?
- If I averaged two zip codes, what place would I get?
- Do equal one-star steps represent equal changes in what is being measured? State your assumption.
- Which column would you sort to find the 'biggest'? Which column has no 'biggest'?

[WARN] LOOK FOR

- 'It has digits, so it's quantitative' — confuses how a value is written with what it measures
- Saying zip code is categorical 'because you can't add them' without saying WHY (it measures/counts nothing)
- Treating Rating as obviously quantitative because the quiz app computes an average star rating
- Naming the individuals as 'the newspaper' or 'the school' instead of the six students

SCORING PART (c) — E / P / I

Expected elements

- **zip-label** (required) — Supports Jordan on ZIP and distinguishes location labels from quantities
- **meaningless-average** (required) — Explains what an average ZIP code fails to measure or describe

- **rating-assumption** (required) — Defends rating as opinion labels, or explicitly assumes an equal-step measured score

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- ‘It has digits, so it’s quantitative’ — confuses how a value is written with what it measures
- Saying zip code is categorical ‘because you can’t add them’ without saying WHY (it measures/counts nothing)
- Treating Rating as obviously quantitative because the quiz app computes an average star rating
- Naming the individuals as ‘the newspaper’ or ‘the school’ instead of the six students

Optional #1 — not collected

DOK 2 For each variable, write **C** (categorical) or **Q** (quantitative); for each Q, give the unit: (1) jersey number of a soccer player; (2) number of goals scored this season; (3) shoe size; (4) preferred position (goalkeeper, defender, midfielder, forward); (5) minutes played in the last match.

Key: (1) C — a label, even though numeric. (2) Q, goals. (3) Q, shoe-size units (a measured foot length on a scale); accept C with a reason that sizes are ordered labels. (4) C. (5) Q, minutes.

[STAR] THE PROBLEM — annotated

A school newspaper surveyed six students for a story about getting to school. The table records lunch period, ZIP code, commute time in minutes, and a 1–5-star opinion rating of the commute.

Two reporters disagree about the numbers in it. **Maya:** “Zip code and Rating are both quantitative — they’re numbers, so we can average them.” **Jordan:** “They’re both categorical. Numbers can be labels.”

Getting-to-school survey (minutes; 1–5 stars)

Name	Lunch	ZIP	Min	Stars
Aiden	A	01902	12	4
Bea	B	01905	35	2
Cruz	A	01902	8	5
Dev	C	01904	22	3
Elise	B	01905	41	1
Fiona	A	01901	15	4

FIRST TAKE — one sentence before you work the parts

Would averaging two ZIP codes tell a reader anything useful about getting to school? Give one reason.

Key: Not graded. Most will say ‘yes, it’s a number’; that is the hook.

[BOOK] VARIABLES (keep on the page)

An **individual** is a person or thing the data describe. A **variable** is a characteristic that can change between individuals. An identifier only names an individual. **Categorical** values are category names or group labels, even when written with digits. **Quantitative** values are numbers for a measured or counted quantity. State units for measurements. Before averaging, ask what quantity the values measure or count.

DOK 1 (a) Who are the individuals in this data set?

Key: The six surveyed students are the individuals.

DOK 2 (b) Classify Lunch, ZIP, and Min as categorical or quantitative. Give the unit for Min.

Key: Lunch and ZIP are categorical labels; Min is quantitative commute time, measured in minutes.

DOK 3 (c) Decide whose ZIP-code claim is right and explain why an average ZIP code would or would not be meaningful. Then defend a classification for the star rating using what the numbers represent.

Key: Jordan is right about ZIP: the digits label places, not measured or counted quantities. Averaging 01902 and 01905 gives 01903.5, which does not describe an average commute or location. The star rating is categorical when its numbers label ordered opinions. Accept a quantitative interpretation only if the student explicitly assumes equal steps on a measured score scale; writing ‘stars’ alone is not enough.